

HVGCD: Rethinking Generalized Category Discovery through Hypothesis–Verification

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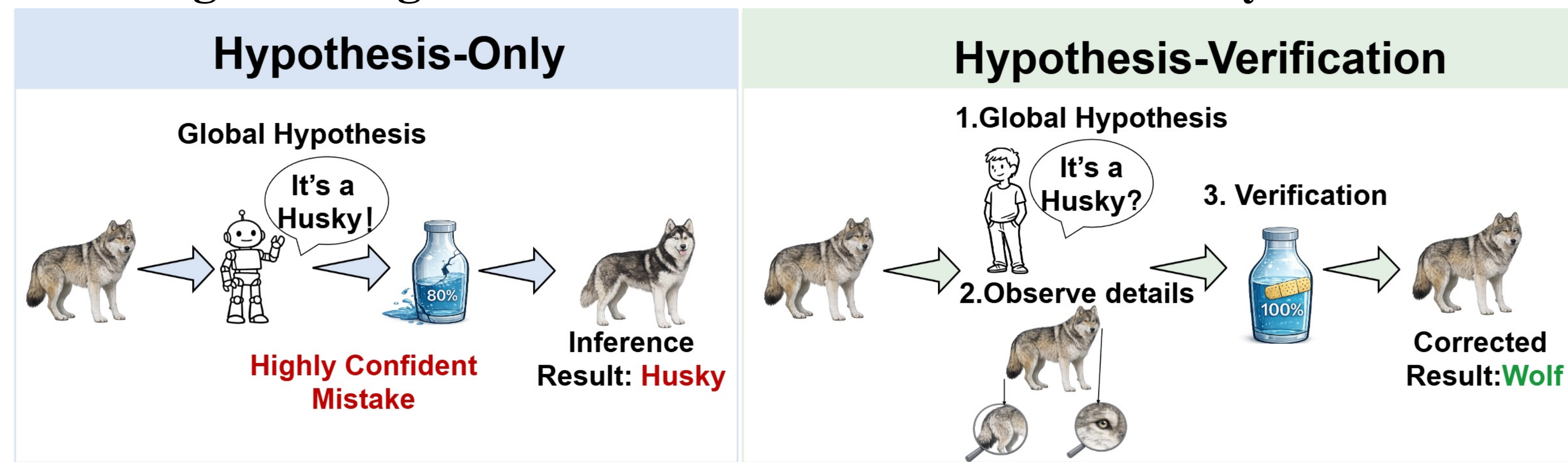


Task & Key Insight

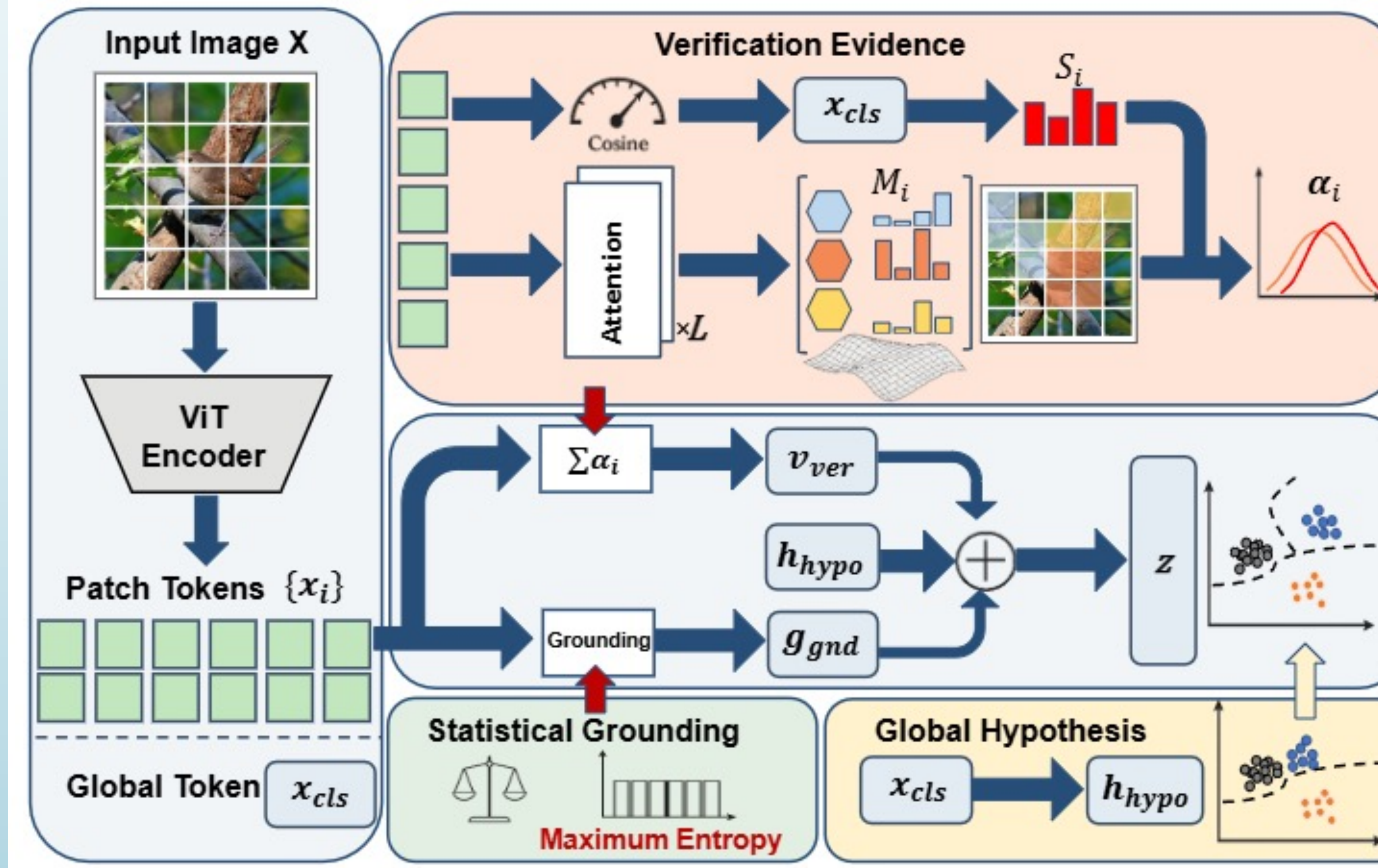
Generalized Category Discovery (GCD) addresses a partially labeled, open-world recognition setting. The goal is to recognize known categories and discover novel categories simultaneously.

Existing GCD methods mainly rely on single-path global embeddings. As a result, unknown samples may be incorrectly assigned to visually similar known categories with high confidence. We formulate GCD as a hypothesis-verification process:

- **Global semantics propose a hypothesis.**
- **Local discriminative cues verify or refine it.**
- **Statistical grounding stabilizes inference under uncertainty.**



Method



HVGCD decomposes representation learning into three complementary components:

Global Hypothesis :

The global class token captures holistic semantic information and provides an initial direction for inference.

Verification Evidence :

Content-adaptive prototypes and dual gating selectively reconstruct discriminative local cues from patch tokens to verify or refine the global hypothesis.

Statistical Grounding:

A maximum-entropy grounding term provides a distribution-level anchor and improves robustness under open-world uncertainty.

The three components are integrated into a unified instance representation.

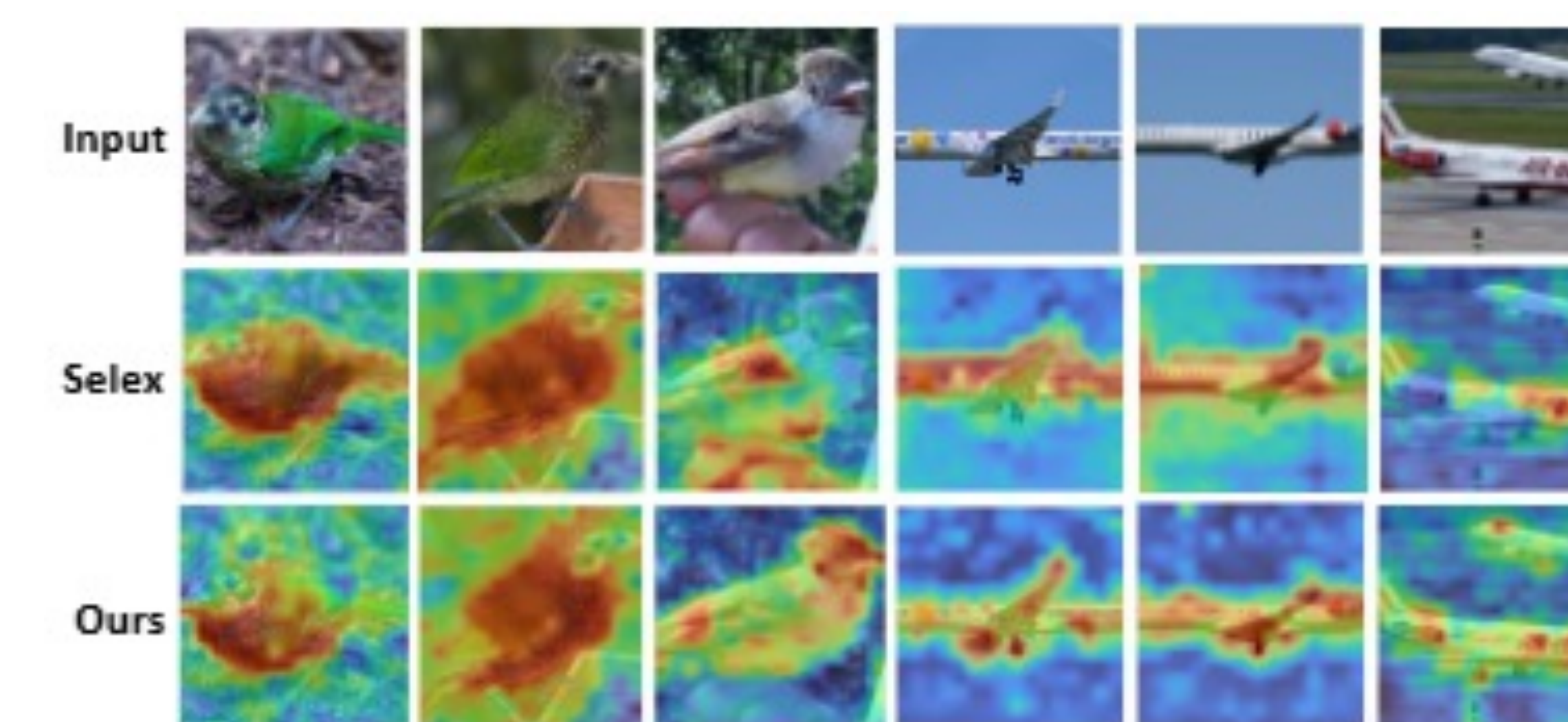
Results

Backbone	Method	CUB-200			FGVC-Aircraft			Stanford-Cars			Herbarium19			Average		
		All	Known	Novel	All	Known	Novel	All	Known	Novel	All	Known	Novel	All	Known	Novel
DINOv1	GCD [54]	51.3	56.6	48.7	45.0	41.1	46.9	39.0	57.6	29.9	35.4	51.0	27.0	42.7	51.6	38.1
	GPC [65]	52.0	55.5	47.5	43.3	40.7	44.8	38.2	58.9	27.4	-	-	-	44.5	51.7	39.9
	XCon [18]	52.1	54.3	51.0	47.7	44.4	49.4	40.5	58.8	31.7	38.1	58.3	27.3	44.6	54.0	39.9
	PromptCAL [64]	62.9	64.4	62.1	52.2	52.2	52.3	50.2	70.1	40.6	37.0	52.0	28.9	50.7	60.0	46.0
	AMEND [11]	64.9	75.6	59.6	52.8	61.8	48.3	56.4	73.3	48.2	44.2	60.5	35.4	54.6	67.8	47.9
	μGCD [56]	65.7	68.0	64.6	53.8	55.4	53.0	56.5	68.1	50.9	-	-	-	58.7	63.8	56.2
	InfoSieve [46]	69.4	77.9	65.2	56.3	63.7	52.5	55.7	74.8	46.4	41.0	55.4	33.2	55.6	68.0	49.3
	SimGCD [63]	60.8	64.8	58.8	52.1	57.7	49.3	55.4	70.9	48.0	43.1	56.3	36.0	52.9	62.4	48.0
	+ Ours	66.5	70.5	64.6	54.1	59.8	51.2	60.6	75.7	53.3	44.6	57.8	37.5	56.4	65.9	51.6
	CMS [12]	67.2	77.0	62.3	52.5	61.0	48.3	51.7	71.2	42.2	35.9	55.0	25.5	51.8	66.1	44.6
	+ Ours	67.9	75.1	64.8	55.1	61.5	52.0	53.9	72.5	44.9	36.4	55.3	26.2	53.3	66.1	47.0
	LegoGCD [6]	61.2	71.4	56.1	50.9	59.1	46.8	57.1	78.8	46.6	44.7	58.3	37.3	53.5	66.9	46.7
	+ Ours	66.9	73.4	63.6	53.8	61.9	49.8	58.8	76.5	50.2	45.8	60.1	38.2	56.3	68.0	50.4
	SelEx [47]	77.0	74.4	78.3	55.2	63.6	51.0	54.5	75.0	44.7	36.0	45.7	30.8	55.7	64.7	51.2
	+ Ours	78.5	76.9	79.3	65.2	68.0	63.8	58.1	75.3	49.8	37.9	48.2	32.3	59.9	67.1	56.3
	Avg. Δ	+3.39	+2.02	+4.20	+4.36	2.45	+5.34	+3.14	+1.05	+4.15	+1.26	+1.52	+1.13	+3.04	+1.76	+3.71
DINOv2	SimGCD [63]	71.1	79.5	66.9	69.1	66.1	70.6	70.9	84.6	64.3	56.2	63.5	52.2	66.8	73.4	63.5
	+ Ours	75.3	82.1	71.9	70.9	72.0	70.3	75.0	85.5	69.9	57.0	65.3	52.5	69.5	76.2	66.2
	CMS [12]	80.1	80.9	79.7	68.9	76.2	64.3	82.8	92.6	78.1	46.4	65.4	36.3	69.6	78.8	64.6
	+ Ours	82.6	84.6	81.7	68.4	75.5	64.9	84.5	93.0	80.4	47.8	65.7	38.2	70.8	79.7	66.3
	SelEx [47]	89.8	87.0	91.2	80.5	78.1	81.6	83.1	93.7	77.0	45.2	55.3	39.7	74.6	78.5	72.4
	+ Ours	90.8	87.3	92.5	81.8	82.1	81.7	83.5	93.9	77.6	46.2	56.1	40.9	75.6	79.8	73.1
	Avg. Δ	+2.55	+2.20	+2.72	+0.87	+3.02	+0.12	+2.05	+0.47	+2.84	+1.06	+0.94	+1.12	+1.63	+1.66	+1.70

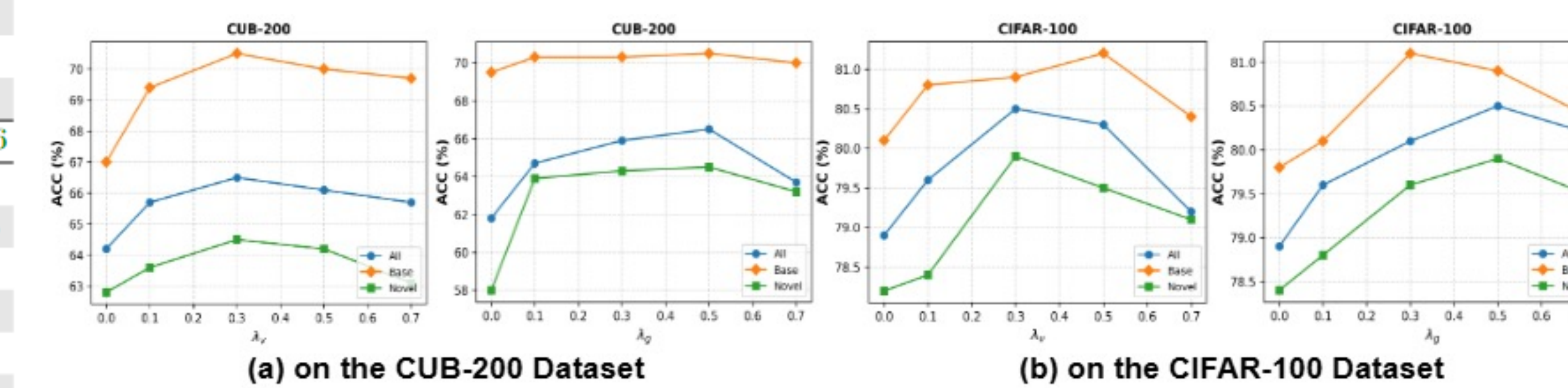
Average improvement of 3.04% on fine-grained benchmarks with DINOv1 features.

Backbone	Method	CIFAR-10			CIFAR-100			ImageNet-100			Average		
		All	Known	Novel	All	Known	Novel	All	Known	Novel	All	Known	Novel
DINOv1	GCD [54]	91.5	97.9	88.2	73.0	76.2	66.5	74.1	89.8	66.3	79.5	88.0	73.7
	GPC [65]	90.6	97.6	87.0	75.4	84.6	60.1	75.3	93.4	66.7	80.4	91.9	71.3
	XCon [18]	96.0	97.3	95.4	74.2	81.2	60.3	77.6	93.5	69.7	82.6	90.7	75.1
	PromptCAL [64]	97.9	96.6	98.5	81.2	84.2	75.3	83.1	92.7	78.3	87.4	91.2	84.0
	PIM [11]	94.7	97.4	93.3	78.3	84.2	66.5	83.1	95.3	77.0	85.4	92.3	78.9
	DCCL [44]	96.3	96.5	96.9	75.3	76.8	70.2	80.5	90.5	76.2	84.0	87.9	81.1
	InfoSieve [46]	94.8	97.7	93.4	78.3	82.2	70.5	80.5	93.8	73.8	84.5	91.2	79.2
	SimGCD [63]	96.9	93.9	98.4	80.5	80.8	79.9	83.4	92.4	78.9	87.0	89.1	85.7
	+ Ours	96.8	92.5	98.9	81.9	82.9	79.9	85.2	93.0	81.2	88.0	89.5	86.7
	CMS [12]	-	-	-	78.0	84.6	64.9	81.8	95.5	75.0	89.9	90.1	70.0
DINOv2	+ Ours	-	-	-	80.2	86.3	67.9	84.5	95.6	78.9	82.4	91.0	73.4
	LegoGCD [6]	97.3	95.8	98.0	79.4	81.9	74.5	84.0	94.9	78.5	86.9	90.9	83.7
	+ Ours	97.5	95.4	98.6	81.2	83.6	76.3	83.4	95.0	77.6	87.4	91.3	84.2
	SelEx [47]	96.3	97.9	95.5	80.9	85.1	72.6	84.0	94.5	78.7	87.1	92.5	82.3
	+ Ours	96.8	97.8	96.2	82.7	86.9	75.3	86.4	95.7	81.8	88.6	93.5	84.4
	Avg. Δ	+0.19	-0.63	+0.08	+1.78	+1.83	+1.90	+1.57	+0.49	+2.10	+1.08	+0.63	+1.36
	SimGCD [63]	98.5	96.0	99.7	86.8	88.6	83.3	89.7	95.1	87.0	91.7	93.3	90.0
	+ Ours	98.6	96.2	99.7	88.3	88.6	87.8	92.4	95.7	90.7	93.1	93.5	92.8
	CMS [12]	-	-	-	90.2	92.5	85.5	93.9	97.2	92.2	92.0	94.9	88.9
	+ Ours	-	-	-	90.6	92.9	85.8	92.2	95.8	90.3	91.4	94.4	88.1
DINOv2	SelEx [47]	98.3	98.7	98.1	87.9	91.1	81.6	90.3	96.5	87.1	92.1	95.4	88.9
	+ Ours	98.2	98.4	98.1	87.8	90.4	82.6	91.7	96.7	89.1	92.5	95.2	90.0
	Avg. Δ	+0.03	+0.05	-0.04	+0.57	-0.12	+1.95	+0.78	-0.23	+1.28	+0.38	-0.17	+1.00

Average improvement of 1.08% on coarse-grained benchmarks with DINOv1 features.



Visualization on attention maps of SoTA and HVGCD



Ablation studies verify the complementary contributions of verification evidence and statistical grounding.

Conclusion

HVGCD restores an explicit verification mechanism to Generalized Category Discovery.

By combining global hypotheses, local verification evidence, and statistical grounding, HVGCD learns more discriminative and robust representations for open-world category discovery.

Extensive experiments on multiple coarse- and fine-grained benchmarks demonstrate the effectiveness and generality of the proposed framework.